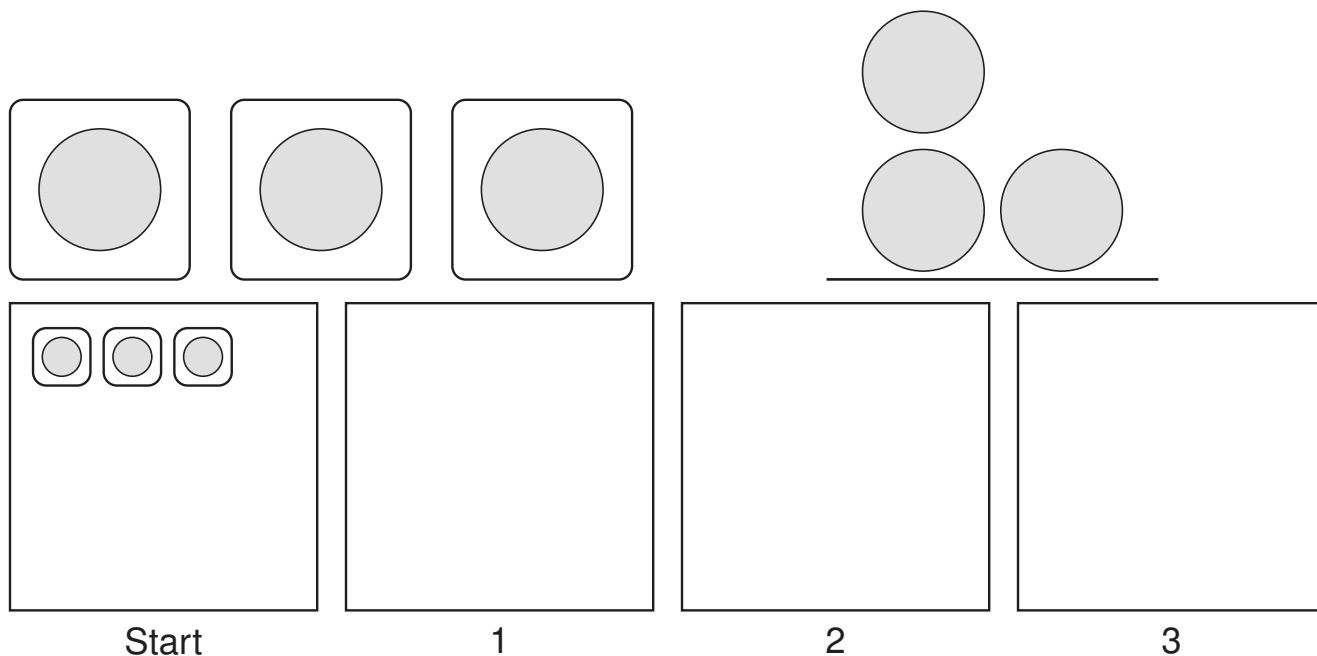
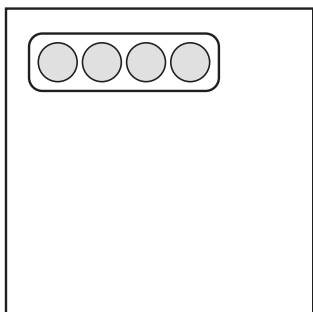
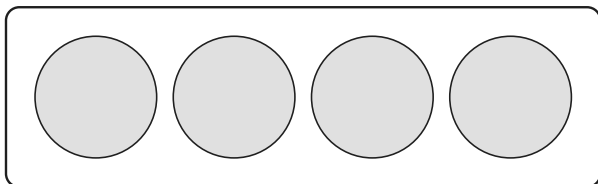


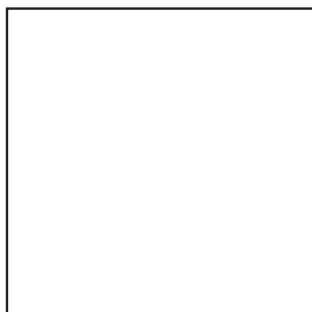
Problem 1: Take 3 counters. Put one counter on each circle in the picture: three piles of 1. A move goes like this: take one counter from every pile, and put those counters together as a new pile. Make one move. Draw your piles in box 1: one dot for each counter, and a ring around each pile. Make another move and draw it in box 2. Make one more move and draw it in box 3. Point to the boxes that show the same piles. Then put your 3 counters on the stairs picture: a tower of 2 next to a tower of 1. Do they fit exactly, one counter on each circle and none left over?



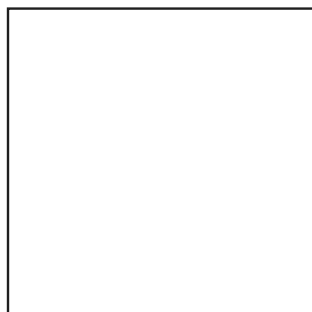
Problem 2: Take 4 counters in one pile. Make 5 moves. Draw your piles after each move. Point to the boxes that show the same piles. Do the piles ever stop changing?



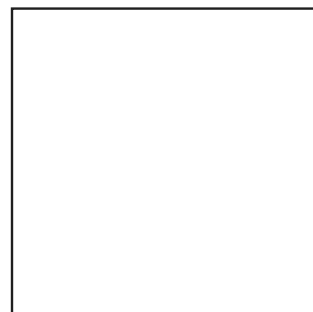
Start



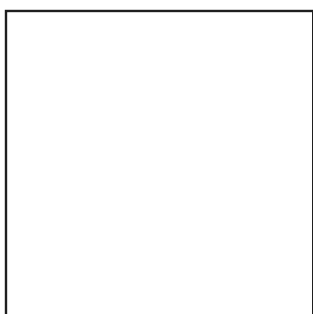
1



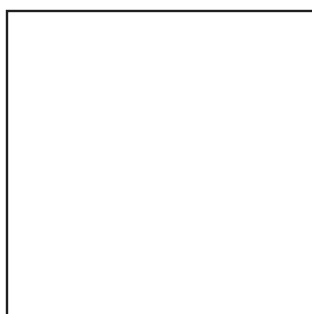
2



3

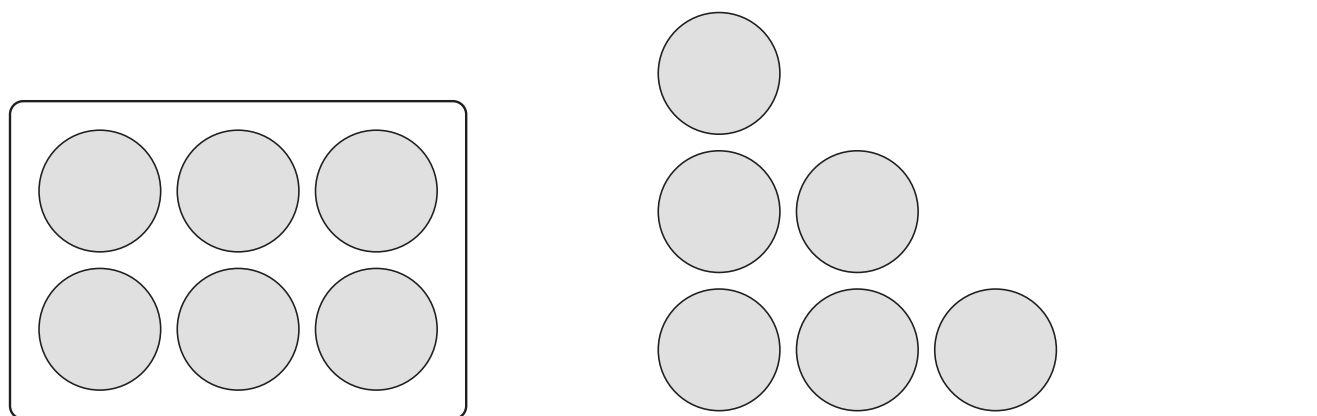


4

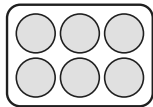


5

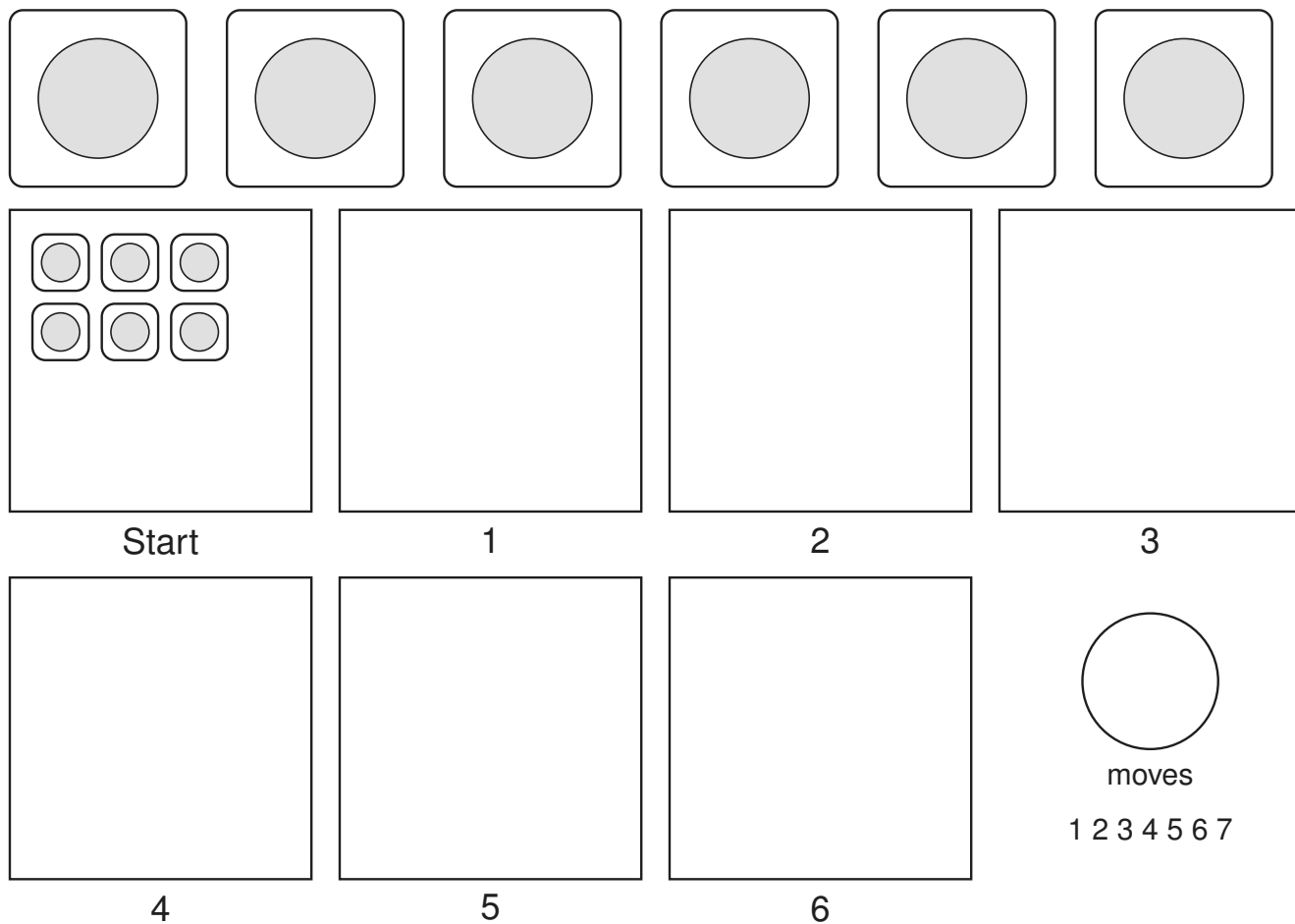
Problem 3: Take 6 counters in one pile. Make moves and draw each one, until a box looks the same as an earlier box. Then put your counters on the stairs picture with towers of 3, 2 and 1. Do they fit exactly?



The diagram shows a box with 6 counters and a stairs picture with 6 towers. The stairs picture has a horizontal line at the base. The first tower has 3 counters, the second has 2, and the third has 1. The remaining three towers are empty.

			
Start	1	2	3
4	5	6	

Problem 4: Put 6 counters in six piles of 1. Make moves and draw each one, until a box looks the same as an earlier box. Did it reach the stairs, towers of 3, 2 and 1? If it did, count the moves it took to get there and write the number in the circle.



The diagram shows the initial state of the game and six boxes for recording moves. The initial state is a 2x3 grid of small circles. Boxes 1 through 6 are empty. To the right of the boxes is a circle for recording the number of moves.

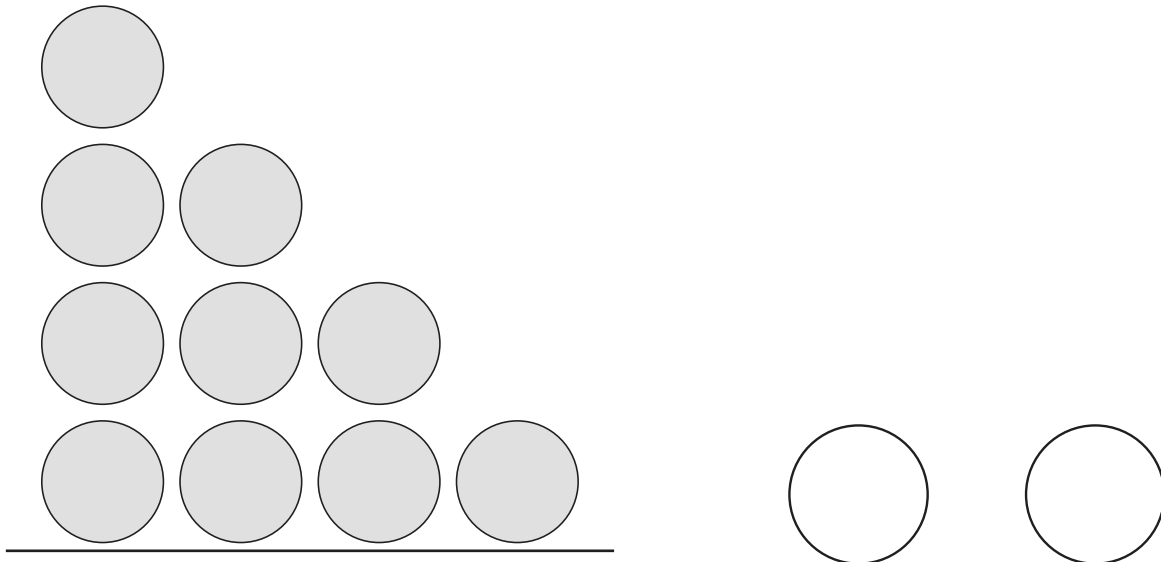
Start 1 2 3

4 5 6

moves

1 2 3 4 5 6 7

Problem 5: Build stairs with 3 counters: a tower of 2 next to a tower of 1. Build stairs with 6 counters: towers of 3, 2 and 1. Now build the next stairs, one step taller, and put them on the big stairs picture. Count the counters. Write the number in the first circle. Then take your stairs apart and build stairs that are one step taller again. Count those counters. Write the number in the second circle.



Problem 6: Take 10 counters in one pile. Make moves, until a box looks the same as an earlier box. After each move, say the sizes of your piles out loud, biggest first. Your grown-up writes the sizes in the box. Then put your counters on the big stairs picture in Problem 5. Do they fit exactly?

Start

1

2

3

4

5

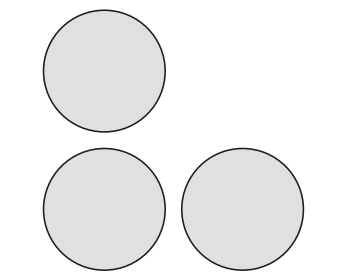
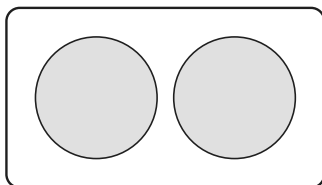
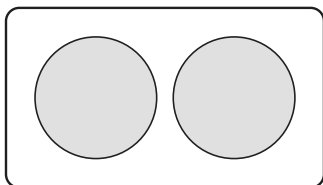
6

7

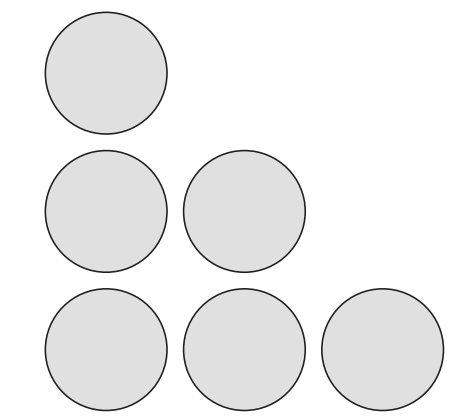
Problem 7: Look back at your games. For each number of counters, make a check mark in the table: did the piles stop changing, or did they go round and round? The last two rows are for other numbers you try.

Counters	Stopped	Round and round
3		
4		
6		
10		

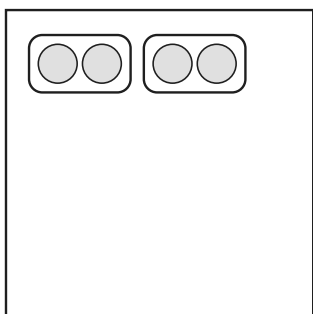
Problem 8: Put 4 counters in two piles of 2. Make 3 moves and draw each one. Which box shows the same piles as the start? Then try to put your 4 counters on the small stairs picture, one on each circle. Try the big stairs picture. Do 4 counters fit exactly on either one?



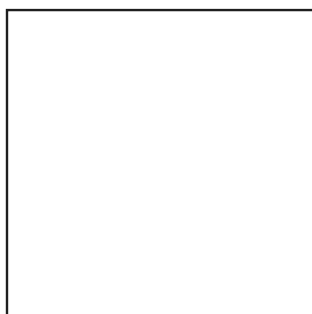
small stairs



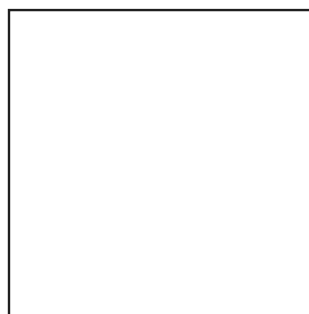
big stairs



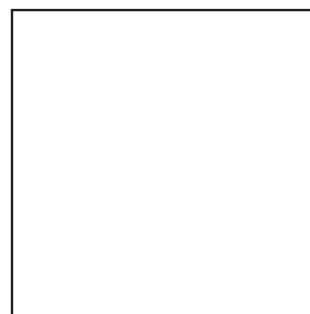
Start



1

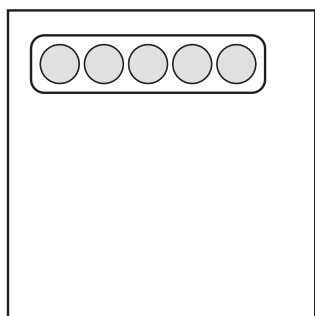
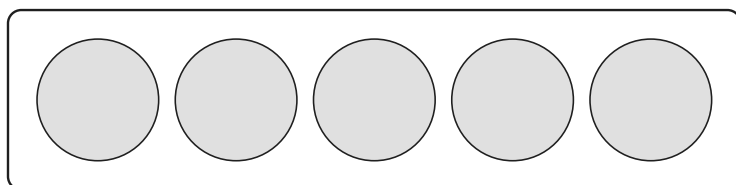


2

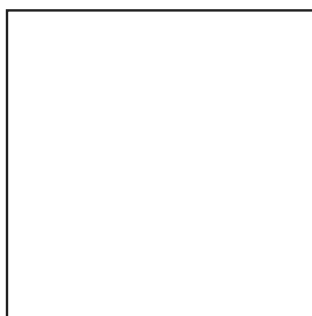


3

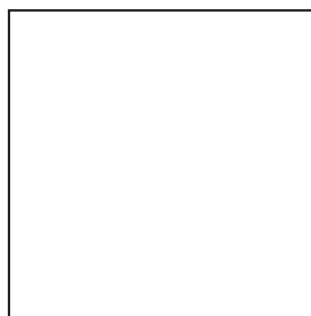
Problem 9: Take 5 counters in one pile. Make 5 moves. Draw your piles after each move. Point to the two boxes that show the same piles. Do the piles ever stop changing? Write 5 in one of the last rows of the table in Problem 7 and make a check mark.



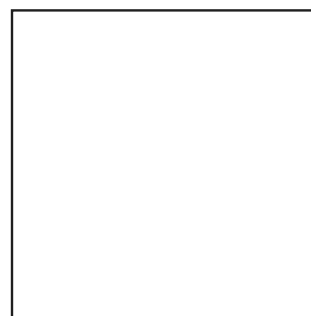
Start



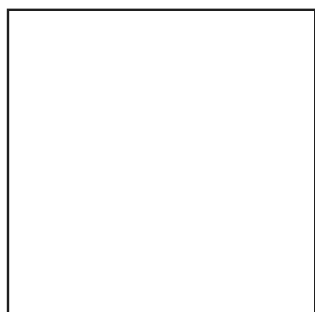
1



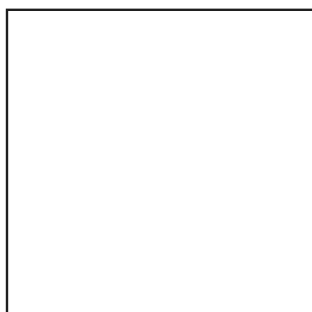
2



3

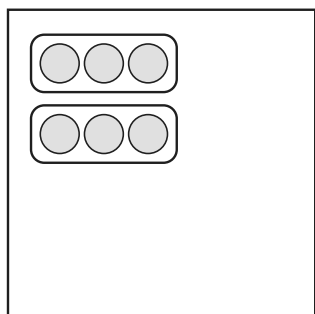
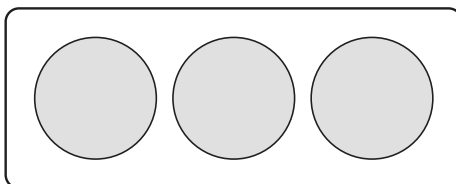
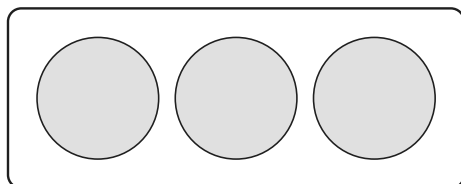


4

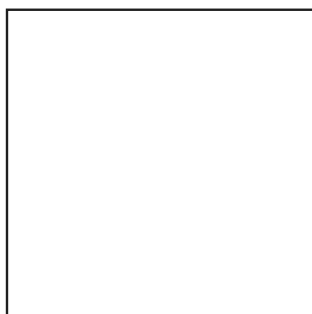


5

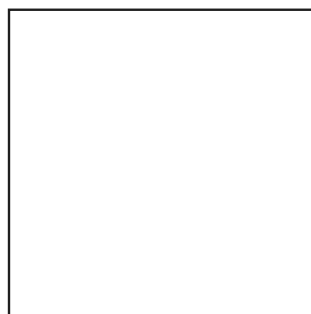
Problem 10: Put 6 counters in two piles of 3. Make moves and draw each one, until a box looks the same as an earlier box. Did it reach the stairs, towers of 3, 2 and 1? If it did, count the moves it took to get there and write the number in the circle.



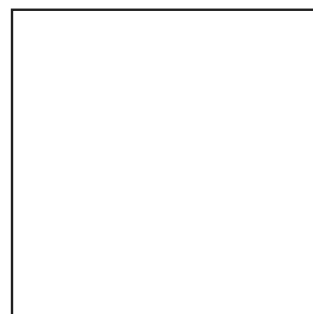
Start



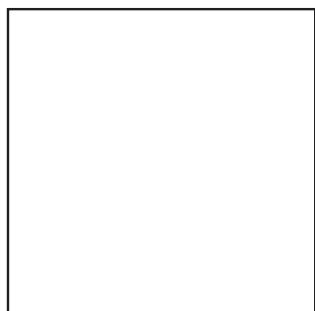
1



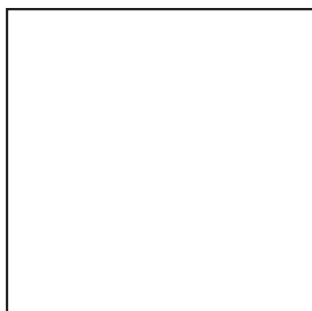
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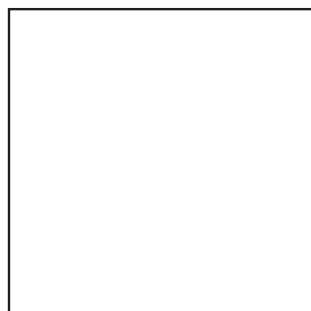
3



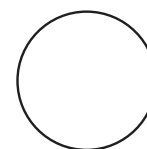
4



5



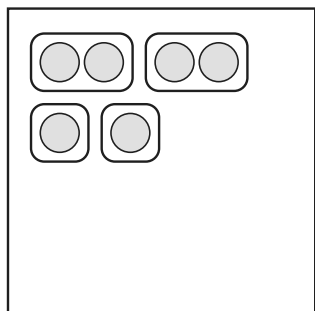
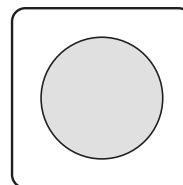
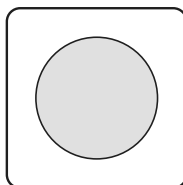
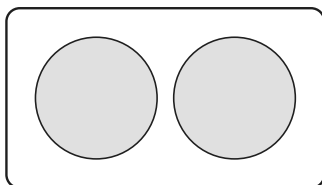
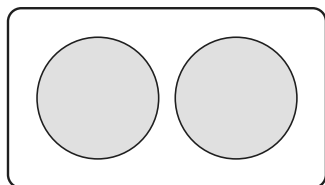
6



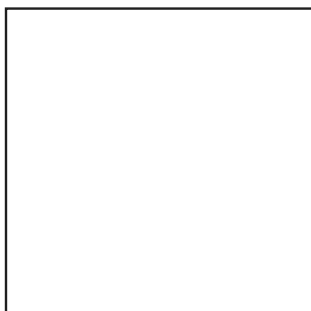
moves

1 2 3 4 5 6 7

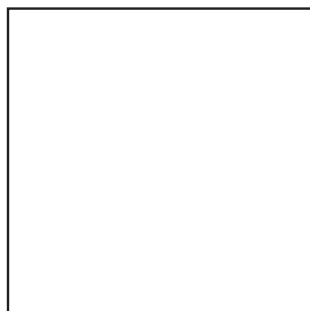
Problem 11: Put 6 counters in piles of 2, 2, 1 and 1. Make moves and draw each one, until a box looks the same as an earlier box. Did it reach the stairs? If it did, count the moves it took to get there and write the number in the circle. Which of your games with 6 counters took the most moves?



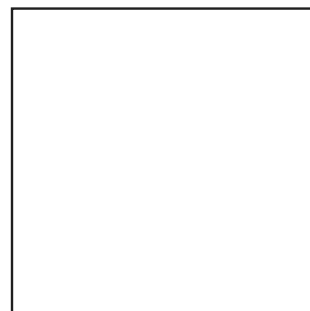
Start



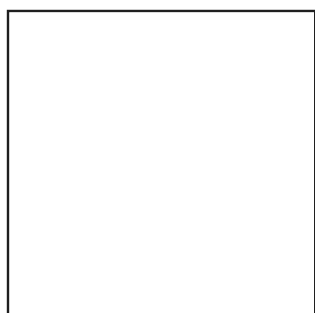
1



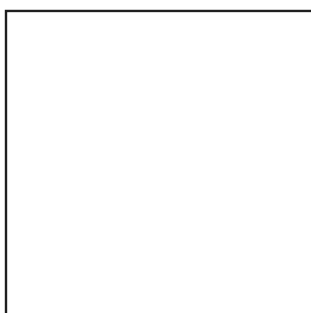
2



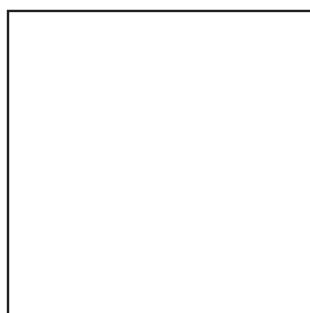
3



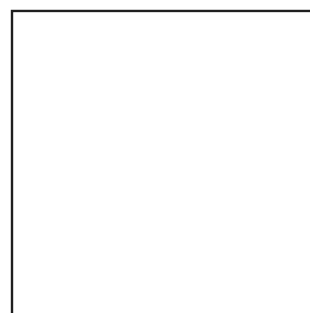
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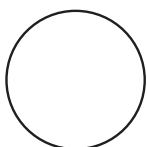
5



6



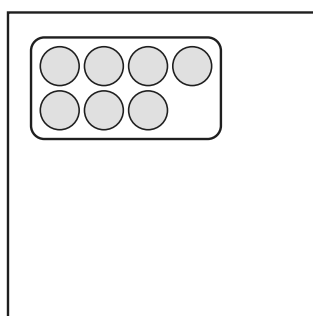
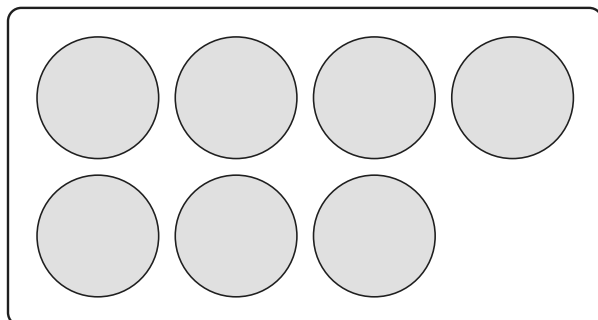
7



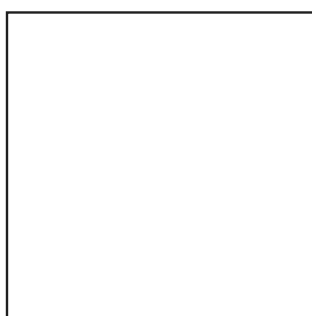
moves

1 2 3 4 5 6 7

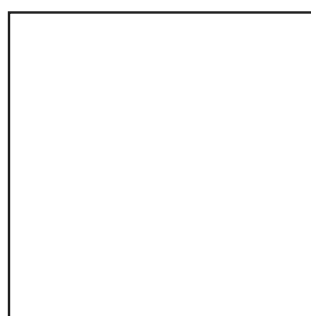
Problem 12: Take 7 counters in one pile. Make 7 moves. After each move, say the sizes of your piles out loud, biggest first. Your grown-up writes the sizes in the box. Point to the boxes that show the same piles as box 3. Do 7 counters stop, or go round and round? Write 7 in one of the last rows of the table in Problem 7 and make a check mark.



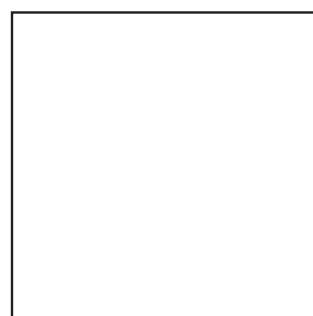
Start



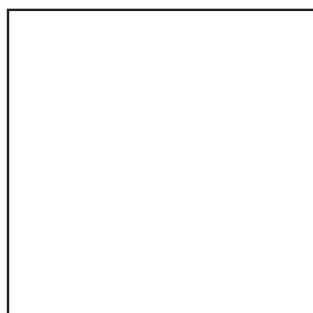
1



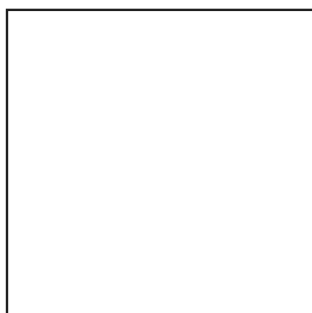
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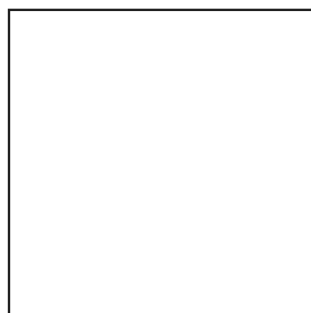
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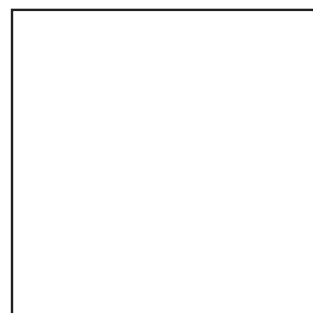
4



5

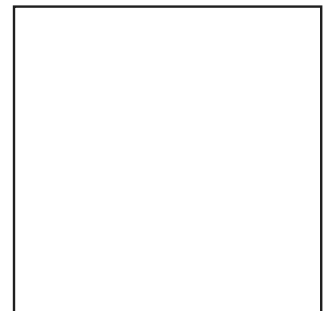
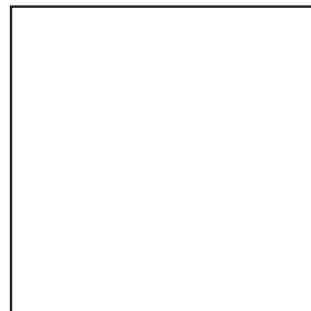
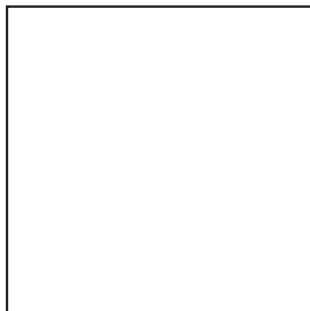
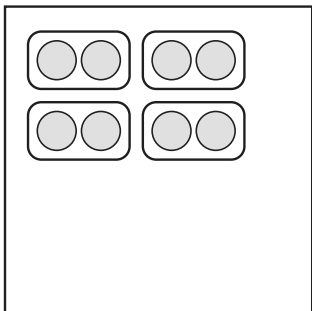
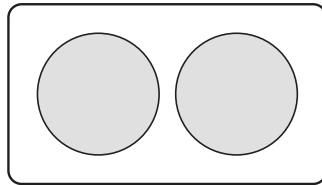
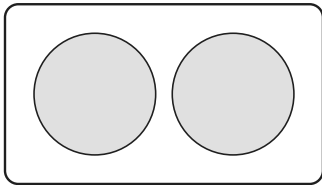
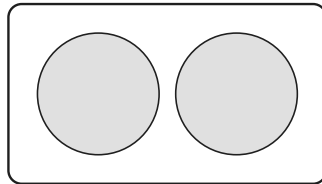
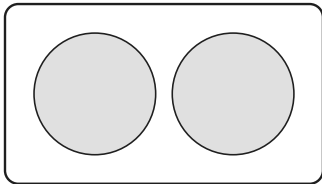


6



7

Problem 13: Put 8 counters in four piles of 2. Make 5 moves. After each move, say the sizes of your piles out loud, biggest first. Your grown-up writes the sizes in the box. Point to the boxes that show the same piles. How many different boxes does this game go round on? Write 8 in one of the last rows of the table in Problem 7 and make a check mark.

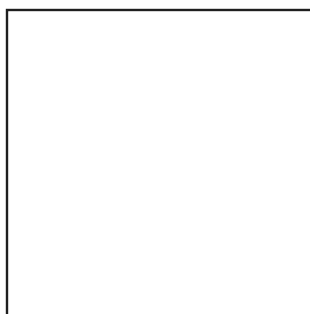
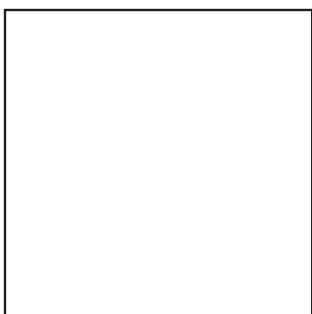


Start

1

2

3



4

5